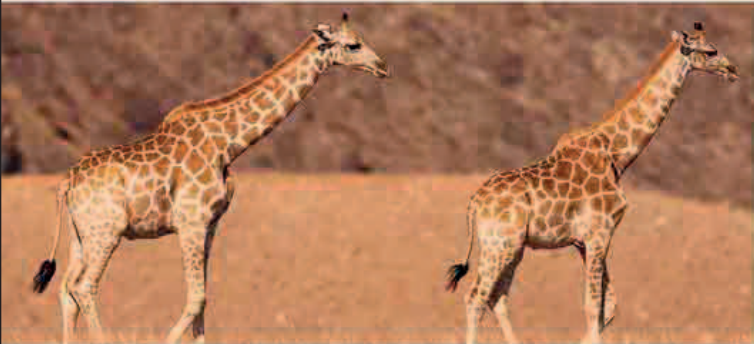




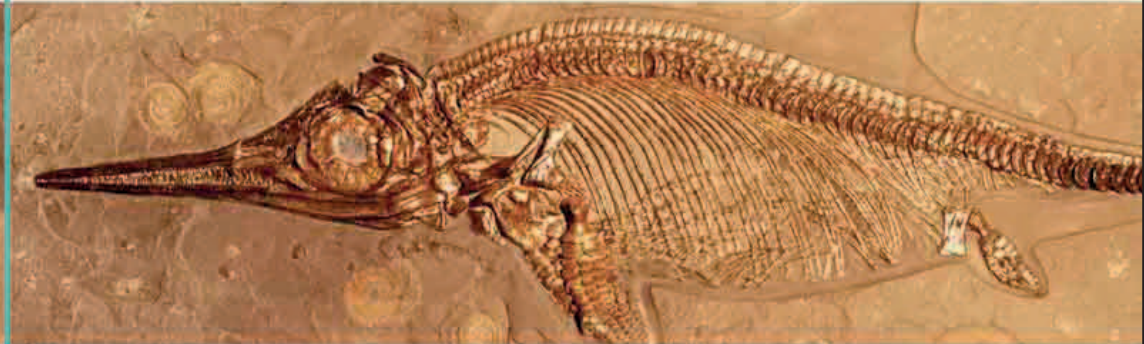
Later commentators often depicted Lamarckian evolution with the idea that giraffes acquired long necks by stretching to reach high branches.

IN 1809, French biologist **Jean de Monet, Chevalier de Lamarck** (1744–1829) came up with one of the first systematic theories about the **evolution of life**. Lamarck argued that life evolved gradually, from the simplest to the most complex. He suggested that a change in the environment can provoke a change in an organism, and that these **changes can also be inherited**. According to him, useful characteristics develop further over the generations, and those that are not useful fall into disuse and may disappear. Unlike Charles Darwin (see 1859), Lamarck had no mechanism to explain how these changes occur. One of his ideas was that an **organism changes during its life to adapt to its environment**, and these changes are then passed on to its offspring. This idea, called **Lamarckian inheritance**, was largely ridiculed by followers of Darwin, but there has been renewed interest with recent discoveries that the

environment can alter genes and their expression—the study of this is known as **epigenetics**.

In Germany, mathematician **Carl Friedrich Gauss** (1777–1855) laid the foundations of astronomical mathematics with his **gravitational constant**. Newton had shown that there is a single universal figure, or constant, for the power of gravitational attraction. Gauss's insight was to devise a simple set of three measurements for calculating **gravity's effects** in which masses are measured in solar masses (the mass of the Sun), distance is measured in terms of the longest diameter of Earth's orbit, and time is measured in days. From these measurements he found a constant for gravity of 0.01720209895. This number was fed into calculations to work out **planetary orbits**. We now know that the measurements Gauss used were not as invariable as originally thought, but despite this his work has been of great value to astronomers.

0.01720209895
THE UNIVERSAL CONSTANT FOR GRAVITY, ACCORDING TO CARL FRIEDRICH GAUSS



Mary Anning's discovery of a fossil, later called ichthyosaur, confirmed that the oceans too once held strange creatures that are no longer alive today.

BRITISH CHEMIST HUMPHRY DAVY (1778–1829) amazed audiences at his London science demonstrations with the glow from the **first electric lamp**, the arc lamp, in which high voltage was shot across the gap between two carbon electrodes. Although bright, the arc lamp was not practical to use for everyday lighting. Everyday electric light arrived only when American inventor Thomas Edison (1847–1931) and British physicist Joseph Swan (1828–1914) developed the incandescent lamp (see 1878–79). Davy also proved that **chlorine is an element**, and that muriatic acid is a compound of hydrogen and chlorine (now known as **hydrochloric acid**), disproving French chemist Antoine-Laurent Lavoisier's theory that every acid contained oxygen.

“ THE CEREBRUM I CONSIDER AS THE GRAND ORGAN BY WHICH THE MIND IS UNITED TO THE BODY. ”

Sir Charles Bell, from *Idea of a New Anatomy of the Brain*, 1811

In 1811, Italian chemist **Amedeo Avogadro** (1776–1856) reconciled John Dalton's atomic theory of elements (see 1803–04) with Gay-Lussac's law of 1808, which said that when two gases react, the volumes of the reactants and products are in simple whole number ratios. Avogadro realized the difference between atoms and molecules. So, simple gases such as hydrogen and oxygen are made of molecules of two or more atoms joined

together. From this, Avogadro deduced his hypothesis that **any gas at the same temperature and pressure** always contains the **same number of molecules**.

Another milestone in chemistry was the system of **chemical symbols and formulae** proposed by Swedish chemist **Jöns Jacob Berzelius** (1779–1848) in 1811 that is still used today. He suggested that every element be identified simply by its initial letter as a capital. Where two elements begin with the same letter, he added a second letter or consonant of the name. To show the number of atoms of an element in a compound, he added a figure to the symbol. So the formula for water is H₂O, indicating there are two atoms of hydrogen for each one of oxygen.

Meanwhile, on the south coast of England, 11-year-old **Mary Anning** made the first of her many key fossil finds. It was of an **ichthyosaur**—a marine reptile

MARY ANNING (1799–1847)



The daughter of a poor cabinet maker in the British coastal town of Lyme Regis, Mary Anning became the greatest fossil hunter of the age. Among her key finds were the almost complete skeletons of marine reptiles such as the ichthyosaur and the plesiosaur. At the time, Anning was one of the foremost experts on the anatomy of these fossil creatures.

French zoologist Jean de Lamarck publishes *Philosophie Zoologique* (Zoological Philosophy) espousing the inheritance of **acquired characteristics**

German mathematician Carl Friedrich Gauss establishes his **gravitational constant**

French physician Philippe Pinel describes **schizophrenia**

1810 British chemist Humphry Davy demonstrates the first electric light: the **arc lamp**

1810 German neuroanatomist **Francis Gall** pioneers the idea of localization of brain functions in particular places in the brain

1810 Humphry Davy shows that **chlorine is an element**

1810 British pharmacist John Haslam describes **paranoid schizophrenia**

1811 Swedish chemist Jöns Jacob Berzelius introduces modern **system of chemical symbols**

1811 British fossil hunter Mary Anning finds an **ichthyosaur** fossil

1811 Italian chemist Amedeo Avogadro's hypothesis